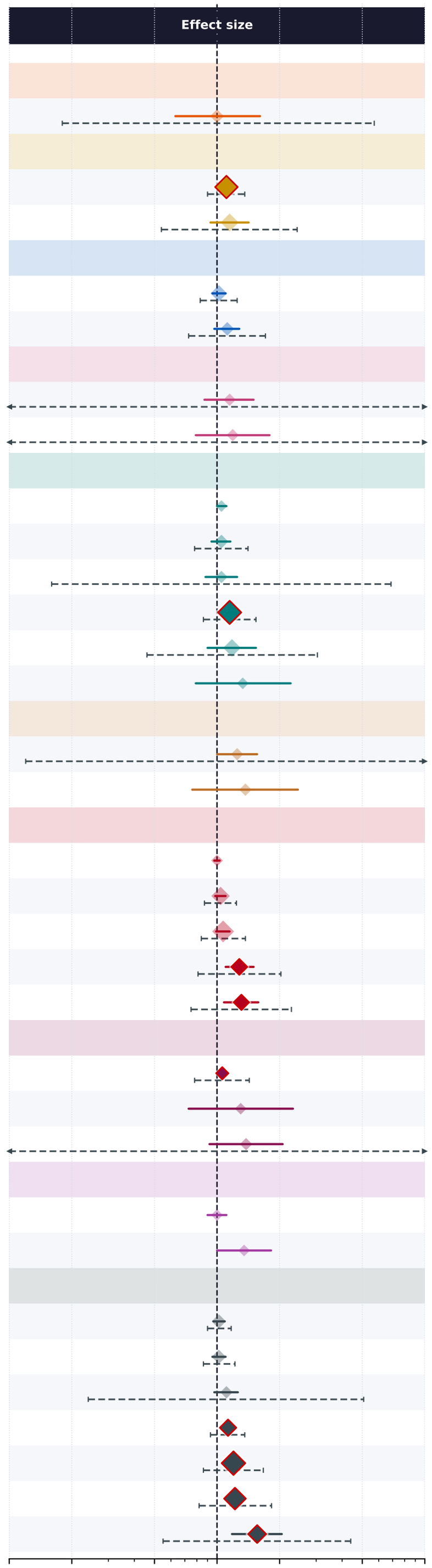
Exposures positively associated with prostate cancer risk

Exposure	Studies	N	Cases	Pooled RR (95% CI)
CAROTENOIDS				
Lutein	5	3,128	1,292	*1.00 (0.63-1.61)*
VITAMINS A, C, D, E, K				
Vitamin D	24	545,005	-	*1.11 (1.04-1.17)*
Vitamin A	13	70,012	17,837	*1.15 (0.93-1.42)*
B VITAMINS				
Folic acid	10	157,607	-	*1.02 (0.95-1.10)*
Vitamin B12	5	5,961	2,681	*1.12 (0.97-1.28)*
ANTIOXIDANTS				
Antioxidants	3	56,601	7,271	*1.15 (0.87-1.50)*
Coenzyme Q10	3	36,416	2,065	*1.19 (0.79-1.79)*
MINERALS & TRACE ELEMENTS				
Copper	2	43,360	6,174	*1.05 (1.00-1.11)*
Iron	5	184,031	11,578	*1.05 (0.94-1.16)*
Magnesium	3	44,056	6,498	*1.05 (0.88-1.25)*
Calcium	26	953,511	-	*1.15 (1.07-1.24)*
Zinc	12	96,718	7,983	*1.18 (0.90-1.54)*
Mineral supplements	2	57,251	1,958	*1.33 (0.79-2.26)*
FERMENTED FOODS & PROBIOTICS				
Yogurt	3	201,324	7,680	*1.25 (1.00-1.56)*
Fermented foods	3	3,460	339	*1.37 (0.76-2.45)*
FATTY ACIDS & LIPIDS				
Docosapentaenoic acid (omega-3)	2	143,344	7,412	*1.00 (0.97-1.03)*
Omega-3 fatty acids	14	867,106	32,787	*1.04 (0.98-1.10)*
Polyunsaturated Fat (PUFA)	22	903,108	-	*1.07 (0.99-1.15)*
Monounsaturated Fat (MUFA)	9	256,816	-	*1.28 (1.10-1.50)*
Alpha-linolenic acid	8	557,816	13,151	*1.31 (1.08-1.58)*
METABOLITES & AMINO ACIDS				
Glucosamine	3	265,129	16,136	*1.06 (1.01-1.11)*
Betaine	2	48,588	868	*1.30 (0.73-2.32)*
Choline	3	8,815	1,545	*1.38 (0.92-2.07)*
DIET				
Mediterranean diet	2	106,146	7,476	*1.00 (0.90-1.11)*
Western pattern diet	2	44,462	1,702	*1.35 (1.00-1.82)*
OTHER				
Multivitamin	6	397,821	7,673	*1.02 (0.96-1.09)*
Ultra-processed food intake	6	248,736	13,760	*1.02 (0.95-1.10)*
Whole-grain intake	3	79,458	7,209	*1.11 (0.97-1.26)*
Processed meat intake	10	1,091,655	65,703	*1.13 (1.05-1.21)*
Alcohol	27	1,849,871	-	*1.20 (1.10-1.31)*
Dairy	22	576,536	30,259	*1.22 (1.11-1.35)*
Red meat	13	926,695	57,504	*1.56 (1.18-2.05)*



Exposure group

Carotenoids · Vitamins A, C, D, E, K · B Vitamins · Antioxidants · Minerals & Trace Elements · Fermented Foods & Probiotics · Fatty Acids & Lipids · Metabolites & Amino Acids · Diet · Other

RR > 1 = positively associated with prostate cancer risk. |◆ pooled RR (size reflects number of studies) |◇ red outline = statistically significant (95% CI excludes 1.0) |◆ faded = not statistically significant | dashed line below CI = 95% prediction interval (where available) | arrowhead = interval continues beyond the axis

Latest publication per cohort

Include subexposures: Yes